TORUS AMPLITUDE OF A QUASI-COHERENT COMPLEX ON A SPACE DESCENDS ALONG FPQC COVERS OF THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 13020-13035 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 13037-13075, the complete author proof (37 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. One typographic slip in the displayed formula (O_{X_, x_i}) is retained verbatim from the author source.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Stalk formulas for tor amplitude on sites, the base-change comparison lemma for derived tensor products (More on Algebra, lemma-base-change-comparison), and flat descent of tor amplitude (lemma-flat-descent-tor-amplitude).

Source reference. Original lemma 0661: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13075–13075. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 068S: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13075–13075. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DJJ: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13075–13075. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DKD: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13075–13075. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04K5: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13075–13075. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $\{g_i : Y_i \to Y\}$ be an fpqc covering of algebraic spaces over S. Let $f : X \to Y$ be a morphism of algebraic spaces and set $X_i = Y_i \times_Y X$ with projections $f_i : X_i \to Y_i$ and $g'_i : X_i \to X$. Let $E \in D_{QCoh}(\mathcal{O}_X)$. Let $a, b \in \mathbf{Z}$. Then the following are equivalent*

(1) *E has tor amplitude in $[a, b]$ as an object of $D(f^{-1}\mathcal{O}_Y)$, and*
(2) *$L(g'_i)^*E$ has tor amplitude in $[a, b]$ as a object of $D(f_i^{-1}\mathcal{O}_{Y_i})$ for all i.*

Also true if "tor amplitude in $[a, b]$" is replaced by "locally finite tor dimension".

Proof. Pullback preserves "tor amplitude in $[a, b]$" by Derived Categories of Spaces, Lemma 0DKD Observe that Y_i and X are tor independent over Y as $Y_i \to Y$ is flat. Let us assume (2) and prove (1). We can compute tor dimension at stalks, see Cohomology on Sites, Lemma 0DJJ and Properties of Spaces, Theorem 04K5. Let $\overline{x}$ be a geometric point of X. Choose an i and a geometric point $\overline{x}_i$ in X_i with image $\overline{x}$ in X. Then

$$(L(g'_i)^*E)_{\overline{x}_i} = E_{\overline{x}} \otimes^{\mathbf{L}}_{\mathcal{O}_{X,\overline{x}}} \mathcal{O}_{X,\overline{x}_i}$$

Let $\overline{y}_i$ in Y_i and $\overline{y}$ in Y be the image of $\overline{x}_i$ and $\overline{x}$. Since X and Y_i are tor independent over Y, we can apply More on Algebra, Lemma 0661 to see that the right hand side of the displayed formula is equal to $E_{\overline{x}} \otimes^{\mathbf{L}}_{\mathcal{O}_{Y,\overline{y}}} \mathcal{O}_{Y_i,\overline{y}_i}$ in $D(\mathcal{O}_{Y_i,\overline{y}_i})$. Since we have assume the tor amplitude of this is in $[a, b]$, we conclude that the tor amplitude of $E_{\overline{x}}$ in $D(\mathcal{O}_{Y,\overline{y}})$ is in $[a, b]$ by More on Algebra, Lemma 068S. Thus (1) follows.

Using some elementary topology the case "locally finite tor dimension" follows too. □